

Cryptography - 9_12

Let's get back to it by recapping LWE remarks.

Remarks:

- Without noise, the problem is easy.
- Error distribution: $\mathcal{X}$ is taken to be any distribution such that $\Pr[|e| > \alpha \cdot q : e \leftarrow \mathcal{X}] \leq \text{negl}(\lambda)$ for $\alpha \ll 1$
- We can combine the m samples in this way: $(A, \vec{b})$ such that $A \in \mathbb{Z}_q^{m \times n}$ and $\vec{b} = \vec{s}^t \cdot A + \vec{e}$ and $\vec{e} \leftarrow \mathcal{X}^m$.
- Decisional version: Distinguish $(A, \vec{b})$ from uniform $(A, \vec{b})$ over $\mathbb{Z}_q^{m \times (n+1)}$. The two are equivalent.

Hardness: For $m = \text{poly}(n)$ and $q < 2$ then $\text{LWE}_{n,q,\mathcal{X},m}$ using $\mathcal{X}$ the the Gaussian distribution with probability $\alpha q \geq 2\sqrt{n}$ ($0 < \alpha < 1$) is at least as hard as GapSVP_γ and SIVP_γ for $\gamma = \tilde{O}(n(\alpha))$.

Now, we take a look at **Regev PKE**.

This is based on LWE:

- $KGen(1^\lambda)$: $\vec{s} \leftarrow \mathbb{Z}_q^n$ is the sk . The pk consists of $m \approx (n+1) \cdot \log q$ samples $(\vec{a}_i, b_i = \langle \vec{s}, \vec{a}_i \rangle + e_i) \in \mathbb{Z}_q^{n+1}$.

We can view them as:

$$A = \begin{bmatrix} \vec{A} \\ \vec{b}^t \end{bmatrix} \in \mathbb{Z}_q^{(n+1) \times m}$$

Note: $(-\vec{s}, 1)^t \cdot A = -\vec{s}^t \cdot A + \vec{b}^t = \vec{e}^t \approx \vec{0} \pmod{q}$.

- $Enc(pk, \mu \in \{0, 1\})$: Pick $\vec{r} \leftarrow \{0, 1\}^m$ and output $\vec{c} = A \cdot \vec{r} + (\vec{0}, \mu \cdot |q/2|) \in \mathbb{Z}_q^{n+1}$.
- $Dec(sk, \vec{c})$: Compute $(-\vec{s}, 1)^t \cdot \vec{c} = (-\vec{s}, 1)^t \cdot A \cdot \vec{r} + \mu \cdot |q/2| = \vec{e}^t \cdot \vec{r} + \mu \cdot |q/2| \approx \mu \cdot |q/2|$.

Correctness: It works as long as $\langle \vec{e}, \vec{r} \rangle$ is less than $\frac{q}{4}$. If $\mathcal{X}$ is a discrete Gaussian w.p. σ then $\langle \vec{e}, \vec{r} \rangle$ has magnitude $\leq \sigma \sqrt{m \frac{\ln(1/\epsilon)}{\Pi}}$ w.p. $\geq 1 - 2\epsilon$.

In particular, we can set $\sigma = (H)(\sqrt{n})$ and $q = \tilde{O}(n)$ which corresponds to $\alpha = \frac{\sigma}{q} = \frac{1}{\tilde{O}(\sqrt{n})}$ and $\gamma = \tilde{O}(n^{3/2})$.

✓ **Theorem**

Assuming LWE is hard, then Regev's PKE is CPA-secure.

We will give a sketch of the proof.